Week 10: Advanced Avellaneda-Stoikov, Structured Products & Finals

Extensions, the corporate view, and communicating results

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

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Contents

Learning Objectives

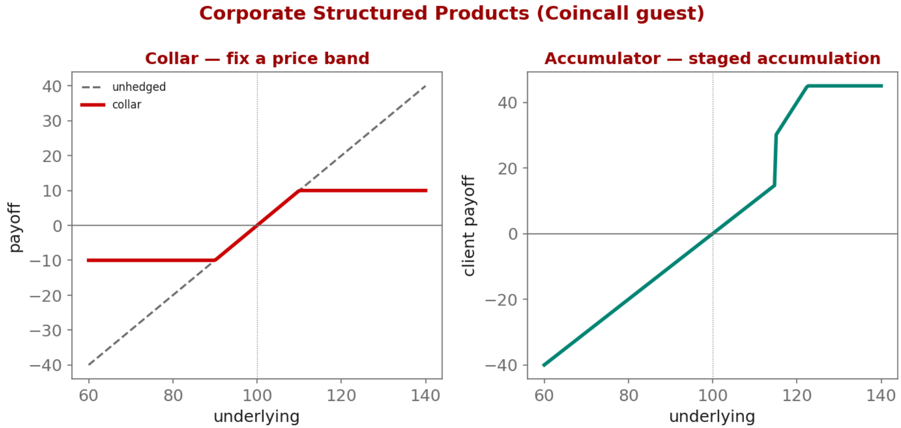
- Implement one advanced Avellaneda-Stoikov extension
- Understand corporate structured products
- Write and present a research-grade report

- Multi-asset market making (BTC + ETH, correlated)
- Adverse-selection-aware and signal-driven quoting
- Robust optimization: worst-case inventory penalty

Lecture 2 (Guest: Fenni Kang, Coincall): Structured Products

- Collars to fix a price band
- Accumulators / decumulators for staged flow
- Coupon / yield-enhancement products
- How that flow lands on the market-maker's book

Illustration



$$\min_{\delta} \max_{\Sigma \in \mathcal{U}} \left\{ \text{spread P\&L} - \tfrac{1}{2} q^{\mathsf{T}} \Sigma q \right\}$$

Why: Robust market making: we do not trust a single covariance estimate Σ, so we optimize against the WORST case over an uncertainty set $\mathcal{U}$. This widens quotes exactly where parameter risk is highest – a principled cure for over-fitting the penalty matrix.

Getting the Data from Coincall

- Multi-asset extension needs BTC and ETH chains together:
  - `GET /open/option/get/v1/BTC` and `GET /open/option/get/v1/ETH`
- Correlation from perp klines (BTC vs ETH) over your backtest window
- Reuse all Milestone-3 Parquet datasets; no new sources required

Cross-check exact paths at https://docs.coincall.com/

Code — week10_collar.py

```
import numpy as np
def collar_payoff(S, spot=100.0, put_K=90.0, call_K=110.0):
    """Long underlying + long put(put_K) + short call(call_K): a fixed band."""
    return (S - spot) + np.maximum(put_K - S, 0.0) - np.maximum(S - call_K, 0.0)

S = np.linspace(60, 140, 9)
print(dict(zip(S.round(0), collar_payoff(S).round(1)))) # capped above, floored below
```

Summary — Key Takeaways

- Pick ONE advanced extension; a clean partial result beats a grand unfinished one
- Robust optimization widens quotes where parameter risk is highest
- Corporate structured products (collars, accumulators, coupons) generate flow
- That structured-product flow lands on – and is hedged by – the maker's book
- Final: report (<= 30 pp.) + 20-minute talk; acknowledge limitations honestly

Glossary — Key Terms

Multi-asset market making jointly quoting correlated underlyings (e.g., BTC and ETH)

Robust optimization optimizing against the worst case over an uncertainty set

Adverse-selection-aware quoting skewing quotes using order-flow signals

Collar long put + short call that fixes a price band around a holding

Accumulator / decumulator products for staged accumulation or liquidation

Coupon product a yield-enhancement structured note

Structured product a packaged derivative payoff sold to clients or corporates

Reading

- Gueant (2016) [chosen extension]; El Aoud-Abergel (2015); +1 paper of your choice

References

- Gueant, O. (2016). The Financial Mathematics of Market Liquidity. Chapman & Hall/CRC.
- El Aoud, S., & Abergel, F. (2015). A stochastic control approach to option market making. Market Microstructure and Liquidity, 1(1), 1550006.
- Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading.

- Plus one paper of your choice relevant to the chosen extension.